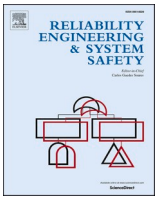
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Multisource information fusion for real-time prediction and multiobjective optimization of large-diameter slurry shield attitude

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ABSTRACT

Abnormal large-diameter slurry shield attitudes (SA) lead to safety and quality problems in shield construction. To achieve intelligent prediction and optimization of the large-diameter slurry SA, a combination of Bayesian optimization categorical boosting (BO-CatBoost) and a multiobjective evolutionary algorithm based on decomposition (MOEAD) is proposed for the prediction and optimization of the tunnel large-diameter slurry SA, as well as the corresponding digital twin framework. Using the Yangtze River large-diameter slurry shield project in Wuhan as an example, data were collected for three soil conditions, namely, silty fine sand, angular gravelly soils and chalky clays, and gravelly soil. The results show that: (1) The R2 values of the three predicted types of soils are all above 0.9, the RMSE values are less than 1.2, and the MAE values are less than 0.9. (2) Based on the results of the SHapley Additive exPlanations (SHAP), the propulsive thrust had the greatest influence on the large-diameter slurry SA. (3) The large-diameter slurry SA optimization improves when the number of adjusted construction parameters increases, with a maximum optimization rate of 23.436%. In addition, a digital twin framework based on BO-CatBoost-MOEAD is proposed, which can effectively control large-diameter slurry SAs and improve overall project safety and quality.

1. Introduction

With the rapid development of the transportation industry, which has prompted tunnel engineering technology to develop toward large cross-sections, large depths and long distances, the application of large section tunnels has become an important direction for domestic traffic construction [1,2]. Slurry-balanced large shields are suitable for environments with abundant groundwater and complex hydrogeological conditions [3]. Therefore, the use of slurry-balanced large shields has been increasingly applied in tunnel construction and buildings in recent years [4], and some scholars have begun to study slurry-balanced large-diameter shields [5,6]. The number of tunnels constructed with slurry shields is also increasing. A large shield has a large palm surface, and due to complex geologic conditions, cutter plate cutting encounters complex geological conditions, increasing the difficulty of controlling the SA [7]. Abnormal SA can result in axis deviation [8], snaking [9], tube sheet extrusion [10], and even shield head destabilization, causing serious safety and quality problems. Therefore, research on large-diameter slurry SAs is especially urgent and important. In actual engineering, large-diameter slurry SAs are mostly adjusted manually by drivers based on experience, which has disadvantages such as lag, and large-diameter slurry SAs must be adjusted in a timely and intelligent manner.

Scholars at home and abroad have performed considerable research on the attitude prediction of large-diameter shields, as shown in Table 1. The methods used to predict large-diameter slurry SAs in the existing research can be categorized into three types: theoretical formula methods, finite element simulations and intelligent algorithms [11]. In the theoretical formula method, SA anomalies in a specific tunnel project are analyzed. The factors affecting the SA of a particular tunnel are analyzed according to the geological conditions, construction schedule, and other conditions of the particular project. Guan et al. [12] developed an inclined shaft shield head adjustment model to study the effects of changes in the tunnel design slope angle on the head adjustment capacity. Shi et al. [13] developed a position–attitude error model

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Nomenclature

SA	Shield attitude
BO	Bayesian Optimization
CatBoost	Categorical Boosting
MOEAD	Multiobjective Evolutionary Algorithm based on Decomposition
SHAP	SHapley Additive exPlanations
BPNN	Back Propagation Neural Network
SVM	Support Vector Machine
RF	Random Forest
XGBoost	EXtreme Gradient Boosting
AR	Advance rate, mm/min X1
CRS	Cutter rotating speed, r/min X2
TT	Total thrust, kN X3
PGAT	Push group A thrust, kN X4
PGBT	Push group B thrust, kN X5
PGCT	Push group C thrust, kN X6
PGDT	Push group D thrust, kN X7
PGET	Push group E thrust, kN X8
PGFT	Push group F thrust, kN X9
SFD	Slurry feed density, t/m^3 X10
SDD	Slurry discharge density, t/m^3 X11
SFFR	Slurry feed flow rate, m^3/h X12
SDFR	Slurry discharge flow rate, m^3/h X13
GPR	Grouting pressure-right, bar X14
GPL	Grouting pressure-left, bar X15
CL	Chamber level, mm X16
CP	Chamber pressure, bar X17
C	Cohesion, kN X18
IFA	Internal friction angle, X19
CHHD	Cutter head horizontal deviation, mm F1
CHVD	Cutter head vertical deviation, mm F2
STHD	Shield tail horizontal deviation, mm F3
STVD	Shield tail vertical deviation, mm F4
PA	Pitch angle, F5
HA	Horizontal angle, F6
MSE	Mean Square Error
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
R^2	Goodness of Fit

Table 1
Summary of previous shield attitude prediction methods.

Method Type	Relevant Reference	Advantages	Limitations
Theoretical Formula Method	Guan et al. [12] Shi et al. [13]	The factors affecting the attitude of the tunnel shield can be analyzed according to the engineering characteristics.	This method has a large degree of subjectivity and cannot be widely applied
Finite Element Simulation	Shen et al. [16] SUN et al. [17]	The adjustment process of SA can be modeled.	The model is relatively simple, and its parameters are difficult to obtain, making it less accurate in practical applications.
Intelligent Algorithms	Liang et al. [18]. Zhang et al. [19] Huang et al. [20] Zhang et al. [21] Wang et al. [22]	Intelligent algorithms can consider various influencing factors more comprehensively and rely on big data for predictions.	When the data volume is overly large, these methods require high amounts of training time and memory.

and derived formulas to calculate the position and attitude errors of a pipe erector grip. However, this approach is mainly based on empirical analysis by technicians, which is highly dependent on their work experience, has a large degree of subjectivity, and cannot be widely applied. In finite element simulation, the forces acting on the shield machine are analyzed, and the adjustment process of the SA is modeled [14,15]. Shen et al. [16] proposed a two-dimensional discrete element model considering the action of an over excavation tool to study the mechanical law of excavation and limit range during SA adjustment. SUN et al. [17] used the finite element method to simulate the rectification process of a shield tunneling machine and concluded that the elastic modulus has a greater influence on the rectification angle and the rectification moment than do other factors. However, this model is relatively simple, and its parameters are difficult to obtain, making it less accurate in practical applications.

The mechanism analysis and finite element simulation methods consider fewer factors, and the parameters of the model are difficult to estimate, leading to low-quality predictions [23,24]. A large amount of data is related to large shields, and these two methods are insufficient for the calculation of large amounts of data. In contrast, intelligent algorithms can consider various influencing factors more comprehensively and rely on big data to predict large-diameter slurry SAs quickly and accurately [25]. Liang et al. [18] used a genetic algorithm (GA) to optimize a back propagation neural network (BPNN), thus improving the prediction of shield tunnel parameters. Zhang et al. [19] proposed merging a differential evolutionary algorithm into an artificial neural network (ANN) to forecast ground settlement during shield tunneling. Huang et al. [20] improved multilayer perception, support vector machine, and gradient boosting regression to predict the tunneling dynamics of shields via a genetic algorithm and principal component analysis. Zhang et al. [21] proposed an algorithm incorporating particle swarm optimization (PSO) and random forest (RF) to optimize the shielding work parameters when the settlement exceeds the tolerance value. Wang et al. [22] constructed an extreme gradient boosting (XGBoost) prediction model and predicted and analyzed the changes in axis deviation using shield construction parameters. The above methods contribute to SA prediction. However, when the data volume is overly large, these methods require large amounts of training time and memory. If the memory is limited, running the method may not be possible, and thus, the SA cannot be predicted and adjusted. Therefore, a more applicable method is needed to quickly process a large amount of data for the efficient and accurate prediction of large SAs.

To reduce prediction error during training, categorical boosting (CatBoost) was proposed by Yandex in 2018; CatBoost enables ordered target statistics and ordered augmentation [26]. CatBoost occupies less memory at runtime, has a shorter training time [26,27], and can better avoid prediction bias than can machine learning methods such as XGBoost, thus improving the veracity of the model [28]. These benefits are complemented by CatBoost's effectiveness with diverse data types, reducing the need for hyperparameter tuning and minimizing overfitting risks for strong generalizability [29]. Since it was first proposed in 2018, CatBoost has achieved success in machine learning studies on large datasets, such as the classification and prediction of data [26]. Yang et al. [30] proposed a novel model mixed sparrow search algorithm and CatBoost, which has better accuracy in credit rating classification

and prediction than CatBoost. Zhang et al. [31] used the difference brain storm optimization algorithm for enhanced precision in transformer fault identification. Kim et al. [32] compared CatBoost with convolutional neural networks (CNNs) and proposed a hybrid CNN-CatBoost model prediction method using temperature, relative humidity and total cloud amount to predict solar radiation. Some civil engineering applications of CatBoost include predicting rock mass qualities [33] and concrete strength [34] and estimating building energy consumption [35]. Therefore, in view of the good performance of CatBoost, a prediction model of SA is constructed based on the CatBoost algorithm to predict the change in SA using various attitude-influencing factors, such as the advance rate and total thrust.

To ensure shield construction safety and obtain more stable large-diameter slurry SAs, multiobjective optimization of large-diameter slurry SAs is needed. Sun et al. [36] proposed a shield tunnel parameter matching model that utilizes a support vector machine (SVM) and an enhanced PSO algorithm to optimize the tunnel parameters. Ge et al. [37] proposed a deep belief network (DBN) approach optimized by the whale optimization algorithm (WOA) for predicting ground subsidence and segmental floatation. These methods optimize the performance metrics of shields very well. However, more than one index exists for expressing large SAs, and multiobjective optimization must be completed. The multiobjective evolutionary algorithm based on decomposition (MOEAD) provides a solution by decomposing the optimization problem into N scalar subproblems and solving them simultaneously through population evolution [38]. Compared to traditional methods such as multiobjective genetic local search (MOGLS) and nondominated sorting genetic algorithm II (NSGA-II), MOEAD exhibits enhanced search efficiency and lower computational complexity, particularly in continuous multiobjective optimization problems [39, 40]. Thus, MOEAD is widely used in classification [41], scheduling [42], and similar operations. Qu et al. [39]. used NSGA-II and MOEAD to transform the Navy's problem of efficient missile boat scheduling into a multiobjective optimization problem and to search for an approximate Pareto-optimal set for the problem. Tan et al. [43] presented a robust optimization algorithm utilizing MOEAD to concurrently optimize both the total distance and the number of vehicles required for transportation. To enhance the effectiveness and success rate of MOEAD in multiobjective optimization, this paper utilizes the nonlinear mapping correlation obtained from the CatBoost prediction model as the fitness function [44]. Therefore, in this paper, an SA intelligent prediction alert and control optimization model is established based on CatBoost-MOEAD.

Additionally, hyperparameter tuning is an important step in machine learning and significantly impacts the performance of machine learning methods. Manually tuning the parameters is an extremely long and complex process. Grid search (GS) and random search (RS) are commonly employed for hyperparameter tuning. Although GS is known for its simplicity and reliability in low-dimensional space, it may not yield satisfactory results in high-dimensional space [45]. RS is simple and easily parallelizable, yet its outcomes are not guaranteed to be optimal. In contrast to both GS and RS, BO outperforms the other methods in terms of optimization performance. Moreover, BO leverages the results obtained from previous tests on hyperparameter values to inform the selection of the next hyperparameter values, thereby enhancing the optimization rate [46]. BO provides clear access to the optimized values of all the hyperparameters, saving time while improving the model accuracy [47]. Sun et al. [48] used the Bayesian algorithm to optimize hyperparameters for logistic regression and RF models, and the accuracy of both models improved after optimization. Shen et al. [49] used a BO approach to optimize the hyperparameters of a deep neural network (DNN) for predicting engine output NOx emissions. Hence, the BO-CatBoost-MOEAD hybrid algorithm is used to construct a large-diameter slurry SA prediction and optimization model.

However, CatBoost can show how much the inputs affect the outputs as a whole but cannot illustrate how changes in the inputs affect the outputs [24,50]. SHAP uses a game-theoretic approach to analyze how different values of each input feature affect the model results [51], which can ensure that the prediction results have good global and local interpretability [52]. Therefore, SHAP is often used in various fields [53-55]. Fabio et al. [51] used SHAP to determine that features related to type of crime, age at first arrest and drug use were most relevant for predicting internal misbehavior. Lin et al. [52] used SHAP to enhance prediction model accuracy by selecting features with significant contributions from the initial feature set. Therefore, in this paper, SHAP is used to analyze the prediction results by specifically assessing the impacts of different construction parameters on large-diameter slurry SAs. This analysis facilitates the effective control of key factors.

This study addresses the following primary research questions: (1) How can a nonlinear relationship between the input parameters and large-diameter slurry SA be established? How can the real-time construction parameters of large-diameter slurry shields be used to predict the attitude of the shield? (2) How can the impact of each input parameter on the large-diameter slurry SA be the key optimization parameter? (3) How can multiobjective optimization of a large SA be achieved and regulated to obtain a stable large-diameter slurry SA with real-time warning and control? Based on these questions, BO-CatBoost is used in this study to create an accurate prediction model for large-diameter slurry SAs, accounting for diverse factors influencing this model. A multiobjective optimization model is constructed based on MOEAD to regulate the deviation of the large-diameter slurry SA.

The contributions of this study are as follows: (1) A combined BO-CatBoost approach is proposed to express the nonlinear mapping correlation between the influencing factors and the prediction target as a fitness function, and real-time shield construction parameters are used to predict the attitude safety of large-diameter slurry shields. (2) SHAP interpretability analysis is utilized to further reveal the key factors influencing large-diameter slurry shields and provides theoretical guidance for improving the attitude control of large-diameter slurry shields. (3) The proposed and developed BO-CatBoost-MOEAD intelligent prediction and optimization method can provide a deeper understanding of the relationships between construction parameters and large-diameter slurry SAs while effectively realizing the multiobjective optimization of large SA control. The number of input parameters can be adjusted in different scenarios to improve the optimization effect. The digital twin framework based on BO-CatBoost-MOEAD proposed in this paper can predict large-diameter slurry SA deviations in advance, providing a basis for staff to adjust the SA in real time.

The structure of this paper is as follows: Section 2 introduces the theoretical foundation of the algorithms used in this paper. Section 3 introduces the methodology of this paper. The prediction and optimization results of the BO-CatBoost-MOEAD models in a large-diameter slurry shield project are analyzed in Section 4. Section 5 discusses the findings. Section 6 summarizes the work and discusses future work.

2. Preliminaries of the method

2.1. BO

BO is a proficient algorithm for globally optimizing parameters in the quest for the extreme value of an unknown objective function, specifically targeting the global optimal solution. The BO framework consists of two basic components: the probabilistic agent model and the collection parameters [56]. BO uses a Gaussian process to train the data and learn the posterior distribution of the objective function due to the highly flexible yet easy-to-operate nature of this type of process. The Gaussian process model is shown in the following equation:

$$P(x) = \frac{1}{(2\pi)^{\frac{n}{2}}|cov|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(x-\mu)^T cov(x-\mu)^{-1} \right) \quad (1)$$

where x is a hyperparameter of the optimization algorithm and n is the

number of hyperparameters. μ (mean) and cov (covariance) are expressed as follows:

$$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i \tag{2}$$

$$cov = \frac{1}{n}\sum_{i=1}^{n} (x_i - \mu)(x_i - \mu)^T \tag{3}$$

The acquisition function is an effective method based on posterior probability distribution, which helps to maximize the potential evaluation points to be uncovered, thus minimizing the total loss function value. The expected improvement is usually chosen as the acquisition function because obtaining the global optimal solution of this function is easy and simple. The expected improvement can be expressed as [57]:

$$EI(X_n) = (y_{best} - \mu(X_n))\varnothing\left(\frac{y_{best} - \mu(X_n)}{\sigma(X_n)}\right) + \sigma(X_n)\varphi\left(\frac{y_{best} - \mu(X_n)}{\sigma(X_n)}\right) \tag{4}$$

where $\varnothing$ signifies the cumulative distribution probability and φ denotes the probability distribution function of the standard Gaussian distribution. Furthermore, y_{best} represents the interim optimum within the current sample space.

2.2. BPNN

With a proper network structure and appropriate weights, a BPNN can effectively approximate any continuous nonlinear function with the desired accuracy. The learning process of a BPNN unfolds through two stages: forward propagation and back propagation. In the forward propagation phase, the actual output is computed from input to output, followed by the correction of weights and thresholds during the back propagation stage, moving from output to input [58]. The output Z_k is shown in Eq. (5) [59].

$$Z_k = f_\varphi\left(\sum_{i=1} W_{jk}\cdot Y_j - \gamma_k\right) \tag{5}$$

where W_{jk} is the weight between the hidden layer unit Y_j and the output layer unit Z_k; γ_k is the threshold value of the output layer neuron Z_k; and f_φ is the activation function of the output layer neuron, where the sigmoid function is usually chosen, i.e., $f(t) = 1/(1 + e^{-t})$.

2.3. SVM

SVM is a machine learning method based on statistical learning theory and structural risk minimization criteria [60,61]. For nonlinear problems, the sample data can be mapped from a low-dimensional space to a high-dimensional space, i.e., the kernel function $\delta(x_i, y_i)$ is used instead of the dot product (x_i, y_i), which makes the samples linearly divisible in the high-dimensional attribute space and then finds the optimal classification plane[61]. The decision function of the SVM is shown in Eq. (6) [62].

$$f(x) = sign\left(\sum_{i=1}^{K} \alpha_i y_i \delta(x_i, y_i) + b\right) \tag{6}$$

where K is the number of support vector machines, α is the Lagrange factor, and b is the bias.

2.4. RF

RF independently samples the training set to produce diverse training sets. Each set contributes to the development of a decision tree, and the collective ensemble forms a random forest used for training and predicting samples [63]. The RF regressor is shown in Eq. (7) [64].

$$f(x) = \frac{1}{N}\sum_{i=1}^{N} f_i(x) \tag{7}$$

where $f(x)$ represents the overall prediction in the RF model, while $f_i(x)$ signifies the prediction of a single decision tree. N is the number of decision trees.

2.5. XGBoost

XGBoost balances the accuracy and complexity of the tree model through the loss function and the regularization term [65], guaranteeing model performance and improving model operation efficiency. The training objective function consists of two elements: the loss from the gradient boosting algorithm and the regularization term, as expressed in Eq. (8) [66].

$$L(\varnothing) = \sum_{i=1}^{n} l(y_i', y_i) + \sum_j \left(\alpha T + \frac{1}{2}\beta \parallel \omega \text{AptCommand2016;}^2\right) \tag{8}$$

where n samples are used for the training function and l is used for the loss of an individual sample. y_i is the predicted value for the ith sample, while y_i' is the true value. N is the number of samples. The parameters α and β are manually set, and ω is a vector composed of the values from all leaf nodes in the decision tree, with T denoting the number of leaf nodes [67].

2.6. CatBoost

Suppose the sample dataset $D = \{(X_k, Y_k)\}_{k=1}^{n}$ and the number of features is m, where $X_k = (x_k^1, x_k^2, \ldots, x_k^m)$ and Y_k is the label value of the target variable. When category features are handled, they are generally expressed as the mean of the label values of the entire dataset, as shown in Eq. (9) [68].

$$\hat{x}_k^i = \frac{\sum_{j=1}^{n}\left[x_j^i = x_k^i\right]\cdot Y_j}{\sum_{j=1}^{n}\left[x_j^i = x_k^i\right]} \tag{9}$$

where x_k^i denotes the i-th fetching value of the k-th training sample. $\hat{x}_k^i$ denotes the value used for replacement.

However, when there are few values or only one fetching value x_k^i of a certain indicator, overfitting often occurs. To prevent overfitting, the samples in D are randomly sorted, and multiple sequences are generated, where any set of random sequences σ, $\sigma = (\sigma_1, \sigma_2, I, \sigma_n)$. Then, the category-type features of each sample are taken and converted into numerical values, which are taken as the means of the labeled values before this sample, after which the prior value P and the weight $\alpha(\alpha > 0)$ of the prior value are added, i.e.,[69].

$$\hat{x}_{\sigma_p}^i = \frac{\sum_{j=1}^{p-1}\left[x_{\sigma_j}^i = x_{\sigma_p}^i\right]\cdot Y_j + \alpha\cdot p}{\sum_{j=1}^{p-1}\left[x_{\sigma_j}^i = x_{\sigma_p}^i\right] + \alpha} \tag{10}$$

where α $(\alpha > 0)$ is the weighting factor and p is usually taken as the average of the target variables in all the data.

2.7. MOEAD

The MOEAD algorithm dissects a multiobjective problem using traditional aggregation methods, breaking it into several single-objective problems [70]. Three commonly used aggregation methods, the weighted summation method (WS), Chebyshev method (TCH) and penalty-based boundary crossing method (PBI), are applied.

Consider $w_1, \ldots, w_N$ as a set of uniformly distributed weight vectors and z^* as the reference point. The approximation problem of the Pareto

Algorithm 1
Bayesian optimization of CatBoost processes.

Input: M_t, N (maximum iterations)
Output: m^*
for $t = 1, 2, \ldots, N$ **do**
  Find m^* by optimizing the acquisition function over the GP: $m^* = argmaxm_m u(m|M_{1:t-1})$
  Get a new sample$(m^*, f(m^*))$
  Augment the data $M_{1:t} = \{M_{1:t-1}, (m^*, f(m^*))\}$ and update the GP
end

Algorithm 2
Normalize the operator in MOEAD with BO-CatBoost.

Input: MOP problems = the result of BO-CatBoost, Terminal condition, N: number of subproblems, T: number of neighborhoods in each subproblem, α: a set of uniformly spread aggregation coefficient vectors, $\alpha = (\alpha^1, \alpha^2, \ldots, \alpha^j, \ldots, \alpha^{Nsol})$
Output: ***EP***: A set that could record nondominated solutions in the searching process
Step 1. Initialization of MOEAD
$EP = \Phi$
Investigate the number of weight vectors T with short distance from α^j
For each $i = 1, \ldots, N$
Set $B(j) = \{j^1, j^2, \ldots, j^T\}$ in the jth subproblem.
Generate initial feasible solutions $\boldsymbol{X} = (x^1, x^2, \ldots, x^j \ldots, x^{Nsol})$ randomly.
For each x^j, calculate the fitness values of m objectives $F_i(x^j)$, $i = 1, 2\ldots, m$
Define the reference point as $z^* = (z^*_1, z^*_2, \ldots, z^*_i \ldots, z^*_m)$ T, where $z^*_i = \min\{F_i(x)|x \in \Omega\}$, $i = 1, 2\ldots, m$.
Step 2. Update the population
Select two **neighbors** x^k and x^l **as parents from** $B(j)$ of the $\boldsymbol{j^{th}}$ **subproblem,** $k, l \in B(j)$, and update them to offspring y using the crossover and mutation mechanism.
Based on the features of the problem, adopt proper operators to repair infeasible solutions, or generate new solutions ***y'***.
Update the reference point for each objective ***i***.
If $z^*_i > F_i(y')$ **then**
$z^*_i = F_i(y'), \forall\ i = 1, 2, \ldots, m$.
end
For each x^j, update neighbor x^p from the set of neighbor B, where $p \in B(j)$
If $g^{me} = (y|\alpha^p, z*) \le g^{me}(x^p|\alpha^p, z^*)$ **then**
x^p is replaced with ***y'***
Else
Preserve x^p
end
After updating the decomposed X, let $\boldsymbol{EP} = \boldsymbol{EP} \cup \boldsymbol{X}$. The relationship is then compared between dominant and nondominated solutions, and ***EP*** is updated.
Step 3. Check the terminal conditions
If $m < Nitr$ **then**
$m = m + 1$ and return to **Step 2**;
Else
Stop the algorithm and output the Pareto set.
end

front can be decomposed into N scalar optimization subproblems via the Chebyshev method, where the objective function of each subproblem can be expressed as follows :

$$g^{te}\left(x\middle|w^j, z^*\right) = \max_{1 \le i \le m}\left\{w_i^j \middle| f_i(x) - z^*\right\} \tag{11}$$

$$w^j = \left\{w_1^j, \ldots, w_m^j\right\}^T \tag{12}$$

MOEAD optimizes the objective functions of all N subproblems simultaneously in a single run. In the MOEAD algorithm, a few close weight vectors are selected in the neighborhood of the weight vectors. The optimal solution of each subproblem forms the population of each generation, and the adjacent subproblems must be used to optimize the current solution.

The pseudocode of BO-CatBoost-MOEAD is as follows (Algorithms 1 and 2):

2.8. SHAP

The SHAP model was introduced in 1953 by Shapley L S [71]. It is a concept of cooperative games and solution values for those games. The SHAP value is a method of assigning the optimal solution for maximizing benefits or exercising the optimal voting rights. The method utilizes the idea of game theory to measure the magnitude of the impacts of features on the model output by calculating the marginal contribution of each feature's SHAP value to the model. The SHAP value may be positive or negative, where a positive value enhances the predicted outcome and a negative value reduces the predicted outcome [72]. Additionally, the importance of a feature increases with both the absolute value of its contribution and its significance in the model [73].

Let the ith sample be x_i, the jth feature of the ith sample be x_{ij}, the predicted value of the model for this sample be y_i, the baseline of the entire model (the mean of the prediction results when the model does not have any input features) be y_{base}, and $f(x_{ij})$ be the SHAP value of the jth feature for the ith sample; then, the SHAP value follows Eq. (13) [74].

$$y_i = y_{base} + f(x_{i1}) + f(x_{i2}) + \ldots + f(x_{ij}) \tag{13}$$

The formula for $f(x_{ij})$ is given as Eq. (14) [75].

$$f(x_{ij}) = \sum_{S \subseteq S(i)} \frac{|S|!(|F| - |S| - 1)!}{|F|!}[v(S \cup \{j\}) - v(S)] \tag{14}$$

where F is the full set of all influences in sample x_i, S is the subset formed by any number of influences in sample x_i, $v(S)$ is the contribution resulting from the joint action of the influences included in subset S, and $v(S \cup \{j\}) - v(S)$ is the contribution of the influences for influence factor j for that joint action.

The mean value of the influence factor j is calculated for all samples to identify the SHAP value of the influence factor j, which is shown in Eq. (15) [76].

$$f(x_j) = \sum_{i=1}^{M} f(x_{ij}) \tag{15}$$

3. Methodology and technical processes

A large-diameter slurry SA prediction and optimization model is established based on BO-CatBoost-MOEAD. This model can achieve highly accurate predictions of large SAs and suggest regulations for bad attitude conditions. First, the evaluation indexes and factors influencing the large-diameter slurry SA are identified to determine the model inputs and outputs, and the data of these indexes are preprocessed. Second, historical data are employed to train the BO-CatBoost prediction model, ensuring precise forecasts of large-diameter slurry SAs. Finally, MOEAD is utilized for multiobjective regulation of the large-diameter slurry SA to obtain a more stable large-diameter slurry SA. Fig. 1 shows the detailed framework flow of this process.

3.1. Data preprocessing

In the digging process, the monitoring system uploads monitoring data in real time at fixed intervals. However, downtime maintenance is sometimes needed in actual projects when detection data are unavailable. Therefore, these abnormal data must be used to improve the accuracy of the model.

When the shield machine is down for maintenance, at least one of the values of the total thrust F, digging speed v, cutter torque T, or advance rate n is 0. Therefore, the product of these four construction parameters is 0 when down for maintenance, while the product is not 0 during normal digging construction. The binary discriminant function is shown in the following equation:

$$D = F{\cdot}T{\cdot}v{\cdot}n \tag{16}$$

$$Status = \begin{cases} normal\ working\ condition, D \neq 0 \\ abnormal\ working\ condition, D = 0 \end{cases} \tag{17}$$

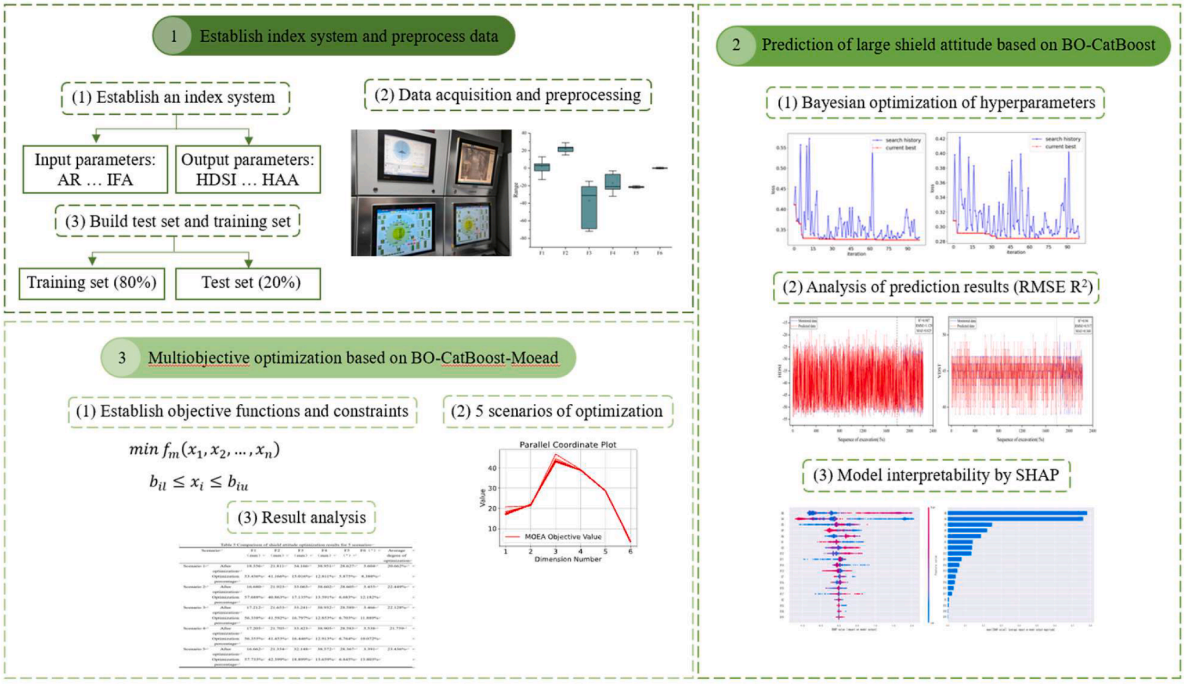


Fig. 1. Framework of large-diameter slurry SA prediction and optimization based on BO-CatBoost-MOEAD.

Electromagnetic noise and strong mechanical vibration at the construction site may affect the monitoring system, resulting in anomalous data that are overly large or small, increasing the error of the model. Therefore, anomalous data that are outside the normal parameter ranges must be excluded to improve the model accuracy.

3.2. Evaluation of the prediction results

To gauge the accuracy of diverse algorithmic models, the root mean square error (*RMSE*), mean absolute error (*MAE*), and goodness of fit (R^2) are used as indexes. The closer the *RMSE* and *MAE* values are to 0, the smaller the error of the prediction results. A higher R^2 value, near 1, reflects an enhanced model fitting effect. These values are calculated as follows:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y^{obs} - y^{pred}\right)^2} \tag{18}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|y^{obs} - y^{pred}\right| \tag{19}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(y^{obs} - y^{pred}\right)^2}{\sum_{i=1}^{n}\left(y^{obs} - \bar{y}^{obs}\right)^2} \tag{20}$$

where y^{obs} is the observed actual value, y^{pred} is the predicted value, $\bar{y}^{obs}$ is the average of the observed actual values, and n is the sample size.

3.3. SA optimization based on BO-CatBoost-MOEAD

3.3.1. Setting the objective function and constraints

BO-CatBoost can accurately predict the large-diameter slurry SA. Therefore, the BO-CatBoost algorithm can be utilized to establish a nonlinear functional connection between large shield construction parameters and attitude. The adaptation function of MOEAD can be obtained according to the BO-CatBoost prediction model, and the objective function of each index of SA is shown in Eq. (21):

$$\min f_m(x_1, x_2, \ldots, x_n) \tag{21}$$

where f_m refers to the objective function and x_n refers to each influential factor.

According to the Shield Method Tunnel Construction and Acceptance Code (GB50446-2017), the attitude of the shield machine must be controlled during shield construction. The permissible deviation limit values are as follows: the permissible deviation limit values for the cutter head horizontal deviation (CHHD), cutter head vertical deviation (CHVD), shield tail horizontal deviation (STHD), and shield tail vertical deviation (STVD) are ±50 mm, and the permissible deviation limit values for the pitch angle (PA) and horizontal angle (HA) are ±30 mm/m. To ensure the safety of shield construction, we choose the values at 80% of the deviation limit values as the early warning values on site. Therefore, during the construction process, the CHHD, CHVD, STHD, and STVD must be maintained within the range of ±40 mm, and the PA and HA must be maintained within the range of ±24 mm/m. Once the deviation exceeds the warning value, timely corrective adjustment is needed.

The constraints of each construction parameter are set according to the specification requirements to obtain reasonable optimization results, as shown in Eq. (22):

$$b_{il} \le x_i \le b_{iu} \tag{22}$$

where x_i represents the large shield construction parameters and b_{il} and b_{iu} represent the lower and upper limits of the constraint conditions, respectively.

3.3.2. Ideal point method

The MOEAD method yields a set of Pareto solutions, each comprising various solution sets. The unique optimal solution can be determined by the ideal point method. The method of calculating the distance between the Pareto scheme and the ideal point for determining the optimal solution given six objectives is shown in Eq. (23) [77].

$$D_n = \sqrt{\sum_{i=1}^{6}\left(\frac{x_{pareto,i} - x_{Epoint,i}}{x_{Epoint,i}}\right)^2} \tag{23}$$

where D_n is the distance from the Pareto solution to the ideal point,



(a) (b)

Fig. 2. (a) Wuhan subway network map; (b) Location map of Ling Wu Village Station to Guobo Center South Station.

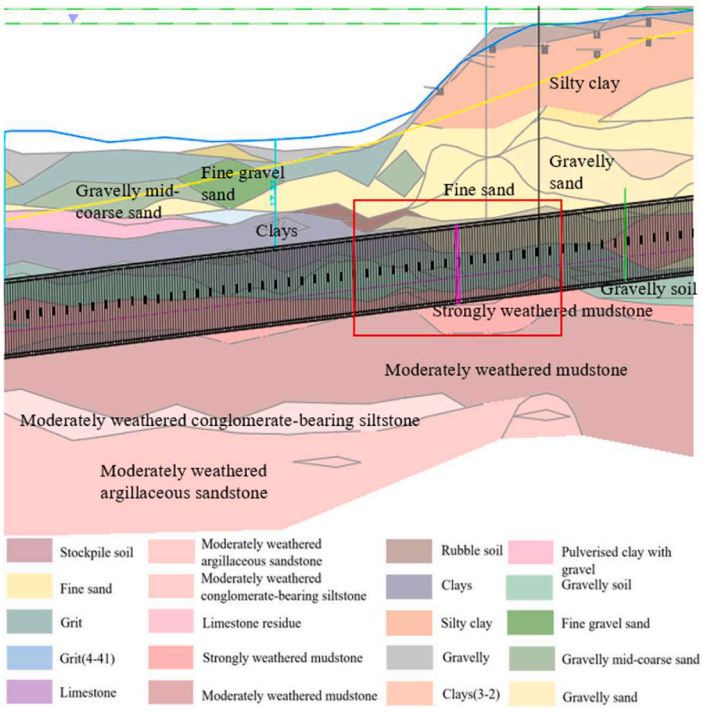
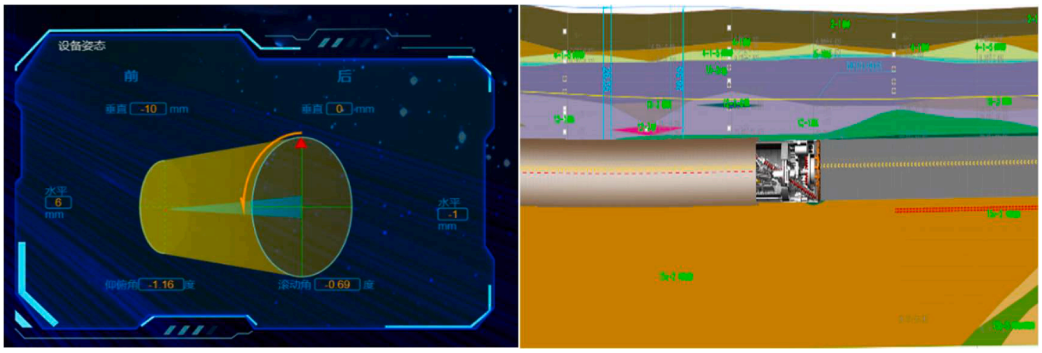


Fig. 3. Localized geological profile.



(a) (b)

Fig. 4. (a) Shield machine attitude monitoring system interface; (b) simulation of shield tunneling.

Table 2
Detailed information on the slurry shield construction parameters.

Parameter type	Variable name	Unit	Maximum	Minimum	Average
Input	Advance rate X1	mm/min	27.39	15.52	22.35
	Cutter rotating speed X2	r/min	0.63	0.58	0.61
	Total thrust X3	kN	86,017.59	60,520.91	70,799.2
	Push group A thrust X4	kN	9553.19	4261.47	5907.15
	Push group B thrust X5	kN	10,626.73	2399.78	6822.59
	Push group C thrust X6	kN	34,814.54	11,362.75	23,019.60
	Push group D thrust X7	kN	17,202.28	9406.52	12,597.97
	Push group E thrust X8	kN	17,203.16	8410.73	11,743.03
	Push group F thrust X9	kN	17,448.79	2474	10,709.05
	Slurry feed density X10	t/m^3	1.33	1	1.22
	Slurry discharge density X11	t/m^3	1.16	1.48	1.32
	Slurry feed flow rate X12	m^3/h	2591.96	2018.5	2318.79
	Slurry discharge flow rate X13	m^3/h	2732.48	2101.04	2419.47
	Grouting pressure-right X14	bar	15.29	5.72	10.77
	Grouting pressure-left X15	bar	9.99	1.17	6.01
	Chamber level X16	mm	0.16	0	0.049
	Chamber pressure X17	bar	3.56	3.98	3.76
	Cohesion X18	kN	30	26	28.00
	Internal friction angle X19	°	16	8	12.01
Output	Cutter head horizontal deviation (CHHD) F1	mm	12	−2	4.26
	Cutter head vertical deviation (CHVD) F2	mm	29	15	21.47
	Shield tail horizontal deviation (STHD) F3	mm	−15	−40	−24.50
	Shield tail vertical deviation (STVD) F4	mm	−10	−32	−22.36
	Pitch angle (PA) F5	°	−20.04	−22.05	−20.96
	Horizontal angle (HA) F6	°	0.75	−0.65	0.08

$\left(x_{pareto}, y_{pareto}, z_{pareto}, w_{pareto}\right)$ is the coordinate corresponding to the optimal Pareto front and $\left(x_{Epoint}, y_{Epoint}, z_{Epoint}, w_{Epoint}\right)$ is the coordinate corresponding to the ideal point.

4. Case study

4.1. Project background

This project site crosses the Yangtze River, totaling a length of 3373.667 m. The shield tunneling interval from Lingwucun Station to Guobo Center South Station is designed as a single-hole, two-lane tunnel with a large-diameter slurry shield machine of 12.56 m in diameter, and the shield tunnel is constructed by the large-diameter slurry shield machine. The tunnel of the interval adopts a single hole and double line arrangement. Each ring of the tube sheet consists of 8 pieces of tube sheet in the form of 5 + 2 + 1. The internal structure of the shield tunnel combines precast and cast-in-place elements and is divided into six parts. Fig. 2(a) shows the subway network map of Wuhan city, and Fig. 2 (b) shows the location map of Lingwucun Station ~ Guobo Center South Station. Buried at depths ranging from 14.11 m to 32.887 m, the tunnel traverses strongly weathered mudstone, moderately weathered mudstone, and fine silt. The localized geological profile is shown in Fig. 3. The surrounding rock and environmental conditions of the construction interval from Lingwucun Station to Guobo Center South Station are more complicated, and controlling the attitude and position of the shield machine is more difficult. Fig. 4(a) shows the interface of the SA monitoring system, and Fig. 4(b) provides a visual of the shield tunneling simulation.

In this paper, CHHD, CHVD, STHD, STVD, PA and HA are selected as the SA prediction targets.

4.2. Input parameter selection and data acquisition and preprocessing

1. Input Parameter Selection

Shield parameters are chosen considering geological conditions, operational characteristics of the shield boring machine, practical engineering experience insights, and relevant references [78]. Increased cutterhead rotation heightens soil disturbance, impacting SA. Total thrust serves as an indirect indicator of interactions between the shield tunneling machine and the excavation surface, with increased thrust leading to increased soil disturbance and consequent effects on SA [22]. Push trust (A–F) is the main execution parameter for the shield operator to correct the SA. Maintaining soil stability and minimizing SA variations can be achieved by judiciously controlling the grouting pressure and volume. The chamber pressure evaluates excavation surface stability and serves as a representative shield machine parameter [79].

Drawing on expert insights, literature [78], and practical engineering experience, an index system encompassing 19 shield input parameters and 6 SA output parameters is proposed, as detailed in Table 2.

2. Data Acquisition

Using Lingwucun Station ~ Guobo Center South Station in Wuhan as the analysis object, the section was constructed by a slurry-balanced shield machine. Real-time data logging and field monitoring facilitated the collection of data under three soil conditions: silty fine sand, angular gravelly soils and chalky clays, and gravelly soil, with approximately 3000 sets of data for each. Using angular gravelly soils and chalky clays as examples, Table 2 lists the detailed information of the obtained data.

3. Data Preprocessing

Due to shield machine downtime for maintenance, sensor signal interference, and other factors, the detected data exhibit many anomalies that must be preprocessed before the data can be used for subsequent prediction and optimization. The outliers in the raw data are eliminated according to Eqs. (16) and (17), as described in Section 3.1. Fig. 5 is a box diagram of the input and output parameters for the angular gravelly soils and chalky clays.

4.3. Model hyperparameter optimization

Among the data obtained from the monitoring, 80% were used for training the model, and the remaining 20% were used as the test set for validation. To avoid overfitting or underfitting, choosing the right combination of hyperparameters is an important prerequisite. There are

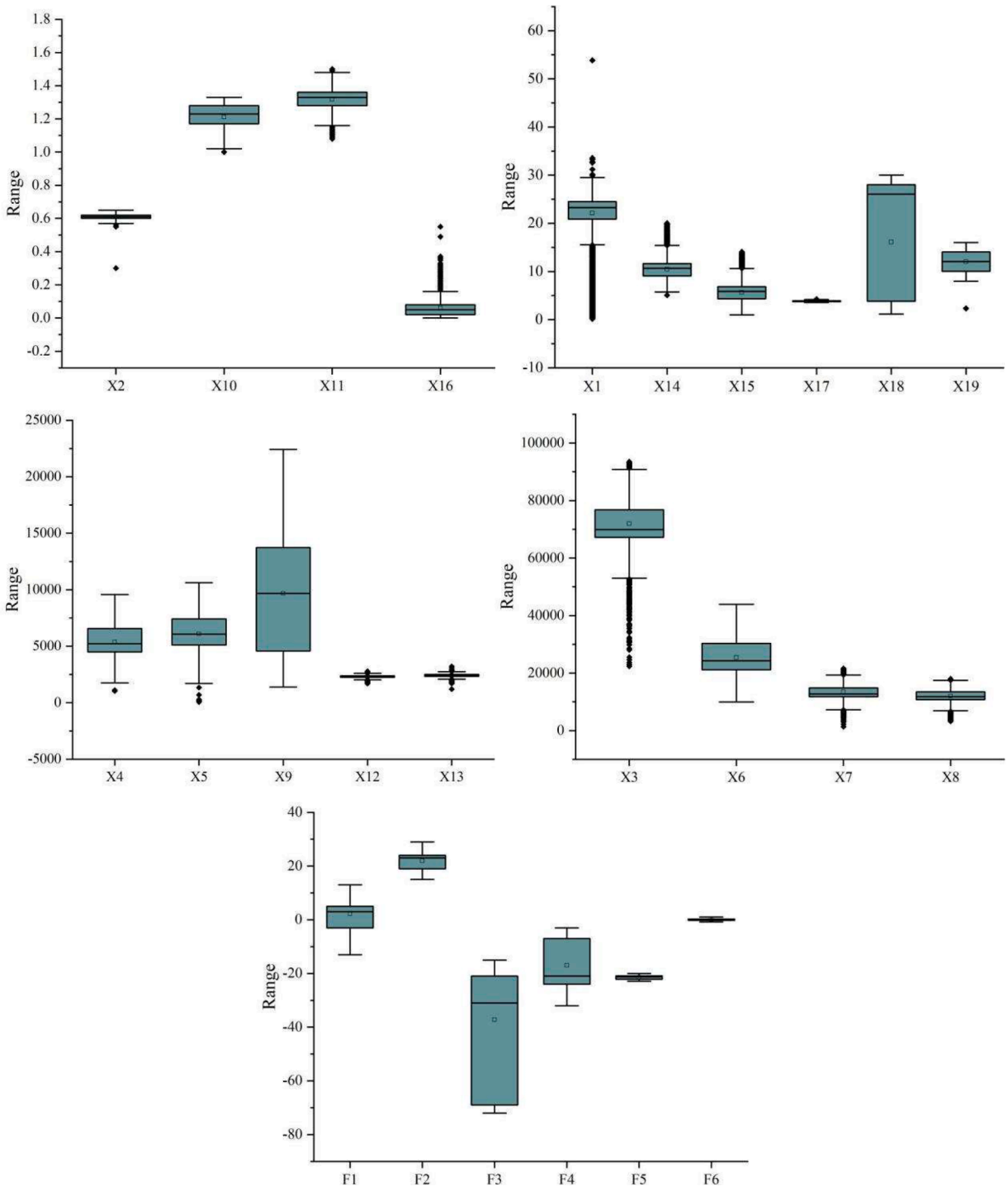


Fig. 5. Input and output parameter distribution.

three main parameters to be regulated in the CatBoost prediction model proposed in this study: the L2 regularization parameter (l2_leaf_reg), the learning rate (learning_rate), and the maximum depth of the tree (max_depth), using Bayesian optimization (BO) for optimization. Identifying the optimal parameters for the CatBoost prediction model involves utilizing the mean square error (MSE) loss function of the training model as the objective function. Before optimizing the parameters, the initial range of hyperparameter optimization needs to be given. The values for l2_leaf_reg, learning_rate and max_depth are set in the initial ranges of [3,9], [0, 0.5] and [4,8], respectively. The hyperparameter optimization results for the CatBoost algorithm are summarized in Table 3. The hyperparametric search process based on BO in gravelly soil is shown in Fig. 6.

4.4. Analysis of the prediction results

The BO-CatBoost prediction model is employed for predicting the SAs for the three soil types. The hyperparameters of the model are set to the optimal hyperparameter combinations, and after sufficient training, the BO-CatBoost model is able to accurately reveal the nonlinear relationship between the input parameters and the SA. Based on the established dataset, the SA prediction results for the three soil types were

Table 3
CatBoost hyperparameter optimization results for different SA prediction targets.

Soil type	Hyperparameter	F1/mm	F2/mm	F3/mm	F4/mm	F5/°	F6/°
Silty fine sand	l2_leaf_reg	3	3	3	3	3	4
	learning_rate	0.0785	0.0878	0.101	0.0599	0.0467	0.0688
	max_depth	6	6	6	8	6	6
Angular gravelly soils and chalky clays	l2_leaf_reg	8	7	4	3	6	5
	learning_rate	0.108	0.117	0.0871	0.031	0.0748	0.0712
	max_depth	8	6	6	6	6	8
Gravelly soil	l2_leaf_reg	5	7	6	4	4	3
	learning_rate	0.109	0.0783	0.114	0.0781	0.0941	0.0475
	max_depth	8	6	4	8	6	8

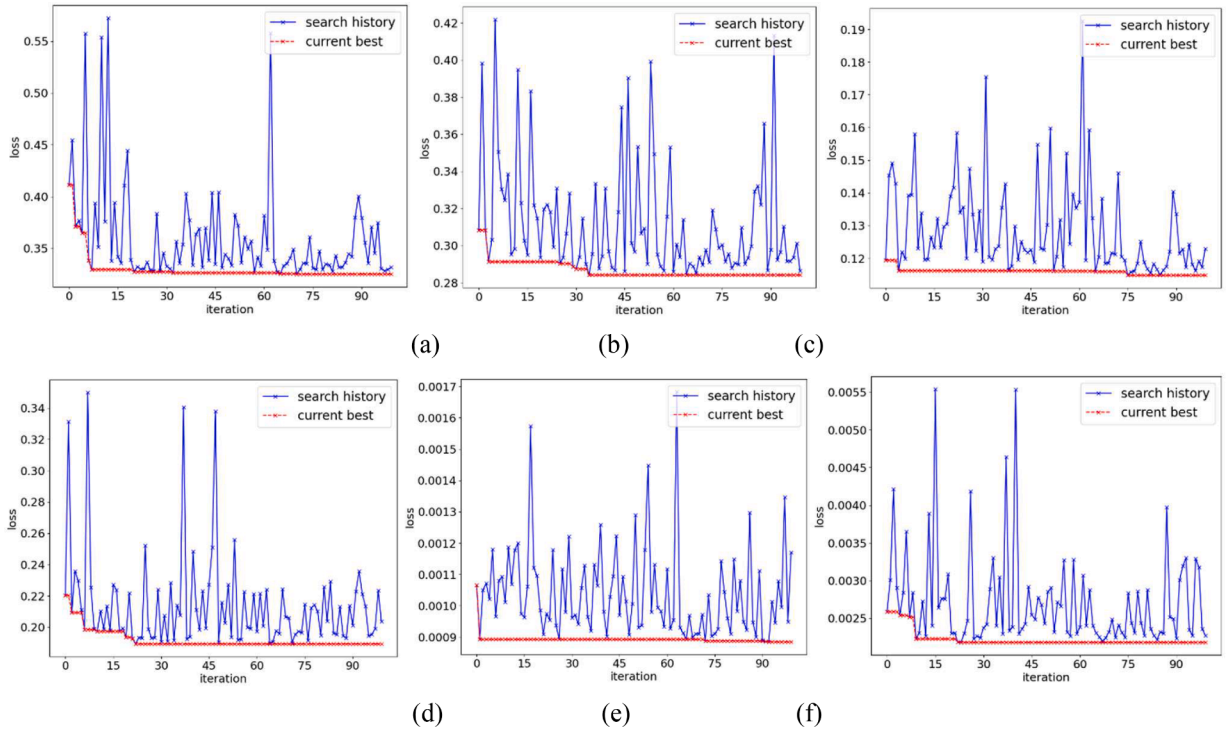


Fig. 6. Hyperparametric search process based on BO: (a) F1; (b) F2; (c) F3; (d) F4; (e) F5; (f) F6.

obtained. Silty fine sand is taken as an example, as shown in Fig. 7. As illustrated in Fig. 7, the BO-CatBoost algorithm effectively learned the correlation between the influencing factors and the large-diameter slurry SA during the training process to predict the large-diameter slurry attitude quickly and accurately. The R^2 values are all greater than 0.9, the RMSE values are all less than 1.2, and the MAE values are all less than 0.9. The BO-CatBoost model was fitted with high accuracy, and the predicted values of the BO-CatBoost model fit the actual values better with less error.

The BO-CatBoost model could also accurately predict the large-diameter SA in the other two soil types. Fig. 8 presents the prediction results for PA and HA in the other two soil types, which shows that the R^2 values obtained for the other two soils range from 0.98 to 0.994, with RMSE values ranging from 0.03 to 0.046 and MAE values ranging from 0.022 to 0.032. The BO-CatBoost prediction model demonstrates advantages of high accuracy and low error in the other two soil types. Therefore, the BO-CatBoost model proficiently forms a nonlinear mapping correlation between the shield construction parameters and the attitude to accurately predict the large-diameter slurry SA.

4.5. SHAP analysis of the predictive models

Prior to the scenario analysis for multiobjective optimization, the extent to which each construction parameter affects the predicted results of the large-diameter slurry SA must be ascertained to determine the parameters to be adjusted during multiobjective optimization. The visual interpretation scheme provided by SHAP can arrange each parameter in the middle line by its importance and label all the samples in the corresponding coordinates according to the Shapley value of each parameter. A negative Shapley value on the left side of the centerline indicates that the parameter has a negative influence on the large-diameter slurry SA prediction model; a positive Shapley value on the right side indicates that the parameter has a positive influence on the large-diameter slurry SA prediction model. In addition, the sample points for each parameter are marked by color, with red indicating large values and blue indicating small values. The combined Shapley value of the parameter and the color of the sample points can be used to analyze the influence of variations in each parameter on the large-diameter slurry SA prediction model. As an example, Fig. 9 shows the SHAP dendrogram of each parameter of the SA in angular gravelly soils and chalky clays.

(1) Propulsion thrust has the greatest effect on the SA. The advance rate, total thrust, slurry feed density, slurry discharge density, slurry feed flow rate and chamber pressure also have greater influences on the SA. Parameters such as the chamber level,

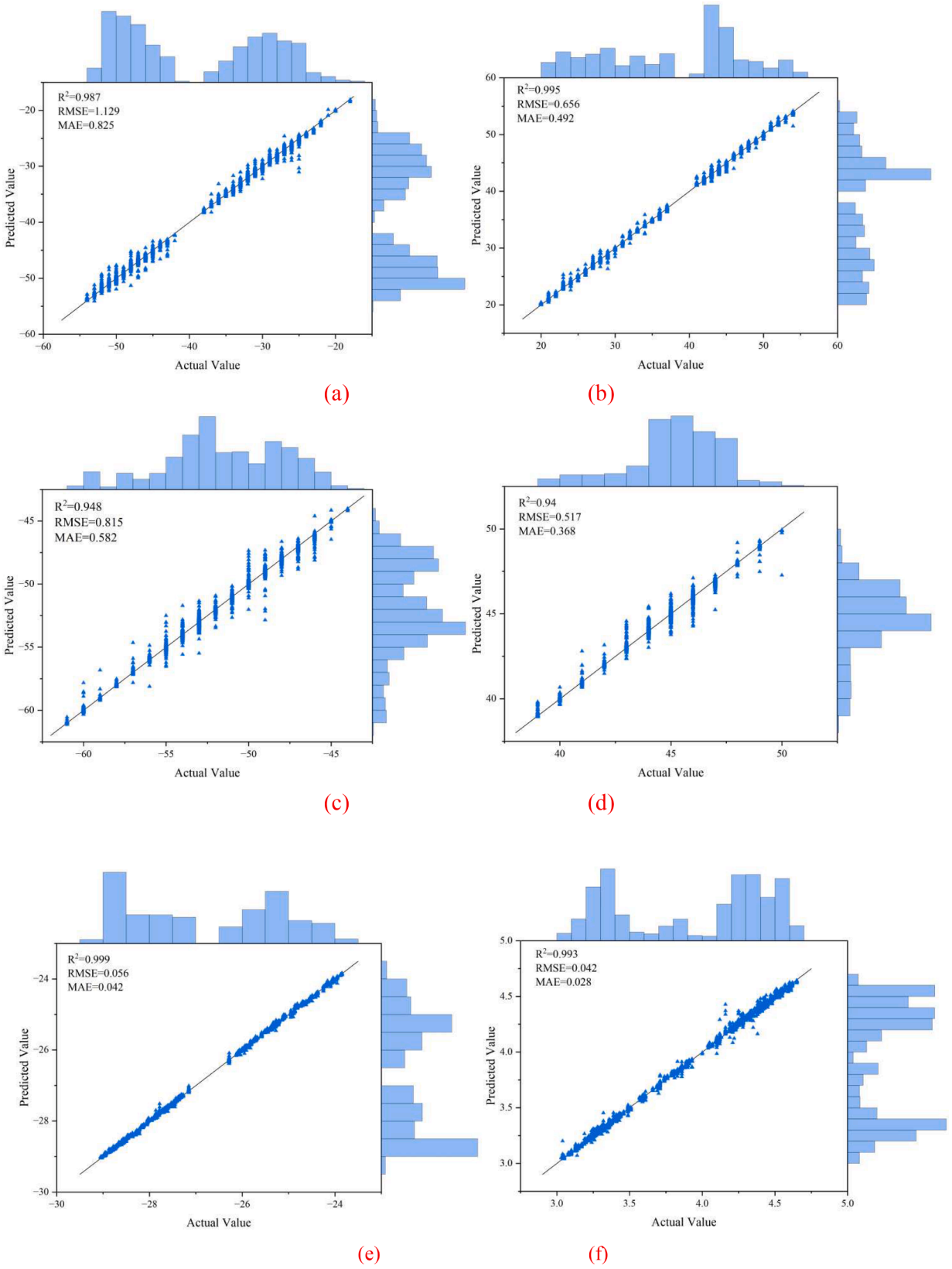


Fig. 7. Prediction results for various indicators in silty fine sand: (a) CHHD; (b) CHVD; (c) STHD; (d) STVD; (e) PA; (f) HA.

cohesion and internal friction angle have less influence. The SHAP values of the six sets of propulsion thrusts in Fig. 9 are larger for all six SA items; thus, these factors exert the most significant impact on the SA. The SHAP values of the advance rate, total thrust, slurry feed density, slurry discharge density, slurry flow rate, and air cushion silo pressure are also relatively large. When the SHAP values of the cutter rotation speed, cohesion and internal friction angle are close to 0, these parameters have

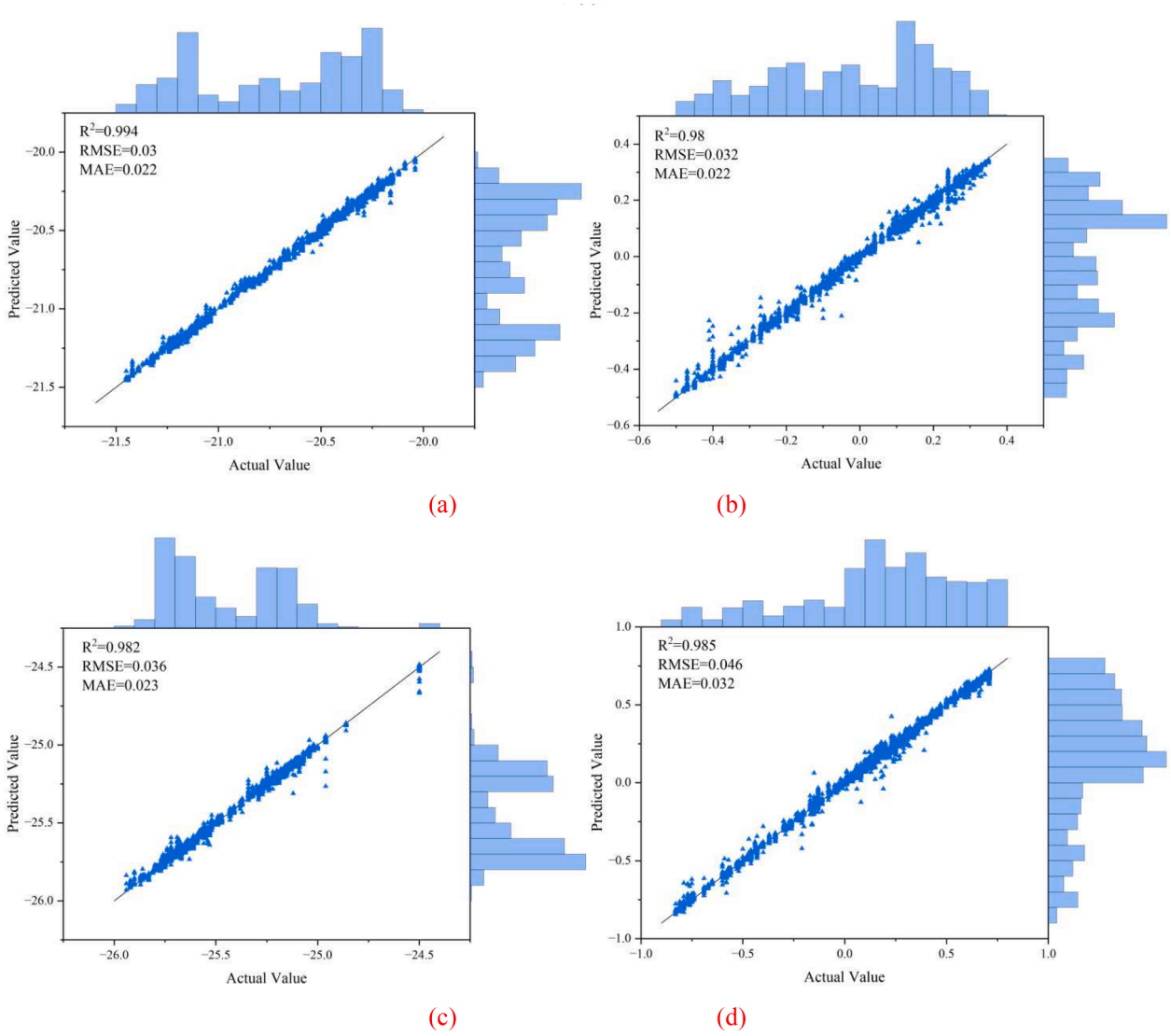


Fig. 8. Prediction results of PA and HA in the other two soil types.

smaller influences on the SA. This is because the soil is categorized into different soils to create a dataset for analysis, and there is very little variation and effect of the soil in the same type of soil.

(2) Thrusts in different directions affect the six objectives of the SA differently. Push group B thrust has positive effects on all six targets. The large-diameter slurry SA deviation increases when the thrust increases. When the thrust of push group A increases, the CHVD and PA also increase, while the CHHD, STHD, STVD and HA decrease. Push group C trust negatively impacts CHHHD and CHVD and positively impacts the other four SA indicators.

4.6. BO-CatBoost-MOEAD multiobjective optimization

There are additional parameters related to the construction of the large shield. To efficiently adjust the construction parameters, the parameters that can be adjusted and have a greater impact on the large SA are selected to optimize the deviation of the attitude of the large shield. The results of the SHAP analysis show that the cutter rotating speed, total thrust, push group A–F thrust, slurry feed density, slurry discharge density, slurry feed flow rate, and chamber pressure significantly impact the SA. To study the influences of different construction parameters and the difference in the number of adjusted parameters on the multiobjective optimization results, five scenarios, as shown in Table 4, are set up by choosing the above parameters.

The SHAP analysis shows that the push group A–F thrust (X4–X9) has the most significant effect on the large-diameter slurry SA, followed by the cutter rotating speed (X2). Therefore, these factors are considered in all scenarios, and these shield construction parameters are the only ones adjusted in Scenario 1. According to the SHAP analysis above, the slurry feed density (X10) and slurry discharge density (X11) are important factors affecting the attitude of large shields. Therefore, these factors are added to Scenario 2. Moreover, combined with the research of other scholars, the impact of total thrust (X3) on the attitude of large shields cannot be ignored. This factor is therefore added to Scenario 3. According to the SHAP analysis results described above, compared with other indicators, the influences of the chamber pressure (X17) and slurry feed flow rate (X12) on the attitude of a large shield are also highly important. Therefore, these two factors are added in sequence in Scenarios 4 and 5. If there is a large difference between the optimized construction parameters and the actual construction parameters, it will cause a hidden danger. To aid construction personnel in adjusting construction parameters, improving their rationality and ensuring shield construction safety, the ranges for large-diameter slurry SA input parameters in silty fine sand are set according to the reasonable parameter range of the shield machine type and relevant norms. In Table 5, the

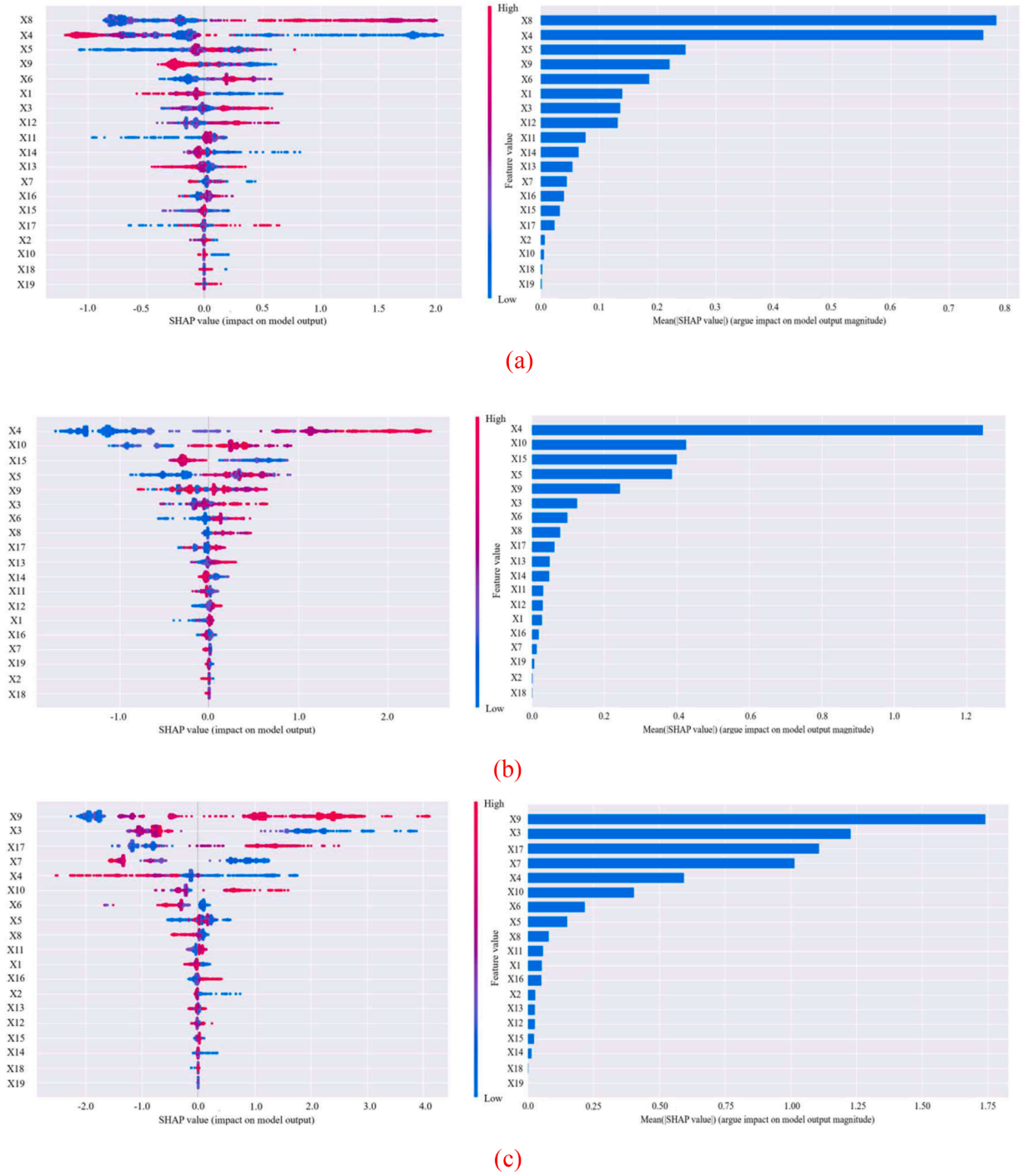
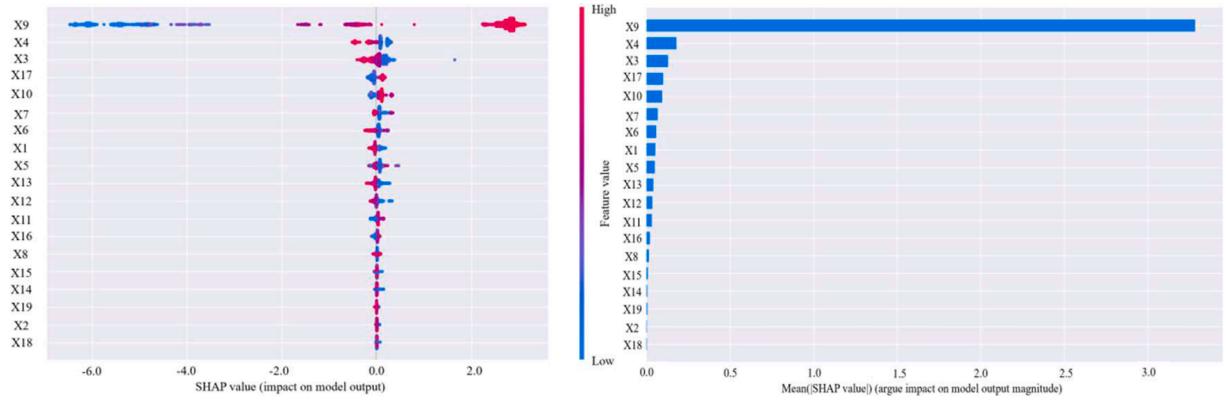


Fig. 9. Dendrogram of the shield parameters for silty fine sand: (a) CHHD; (b) CHVD; (c) STHD; (d) STVD; (e) PA; (f) HA.

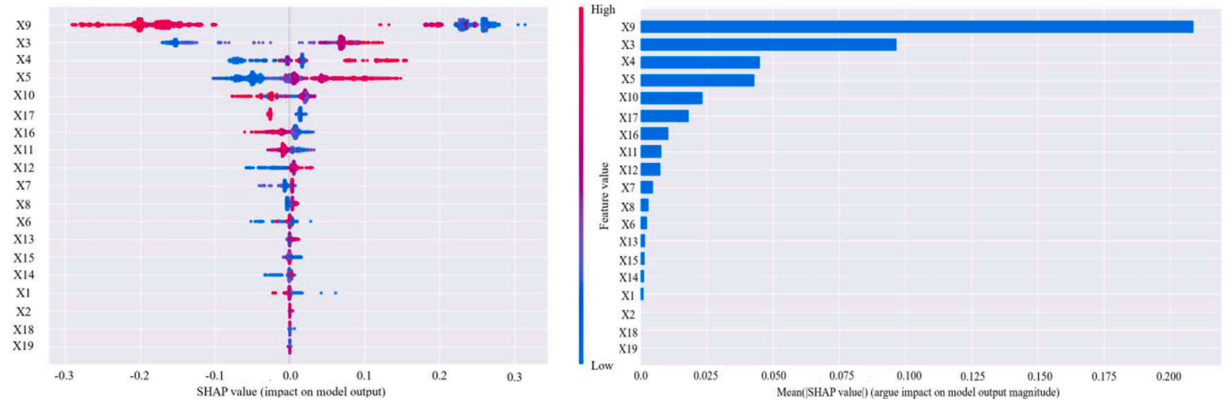
input parameters for the optimization of the large-diameter slurry SA of silty fine sand are all within the set initial reasonable constraint range.

The BO-CatBoost-MOEAD method is used to optimize six objectives—CHHD, CHVD, STHD, STVD, PA and HA—in the scenarios. To obtain the optimal parameters of MOEAD, the relevant algorithm parameters are set after preliminary analysis as follows: the maximum number of evolutionary generations is 100, and the population size is 100. Table 6 compares the optimization rates of the Pareto frontier solution sets for each output in the five scenarios. To achieve the best optimization for the large-diameter slurry SA, the ideal point hair is utilized to pinpoint the most favorable choice within the Pareto solution set. The best optimization rates for different scenarios are shown in Table 6.

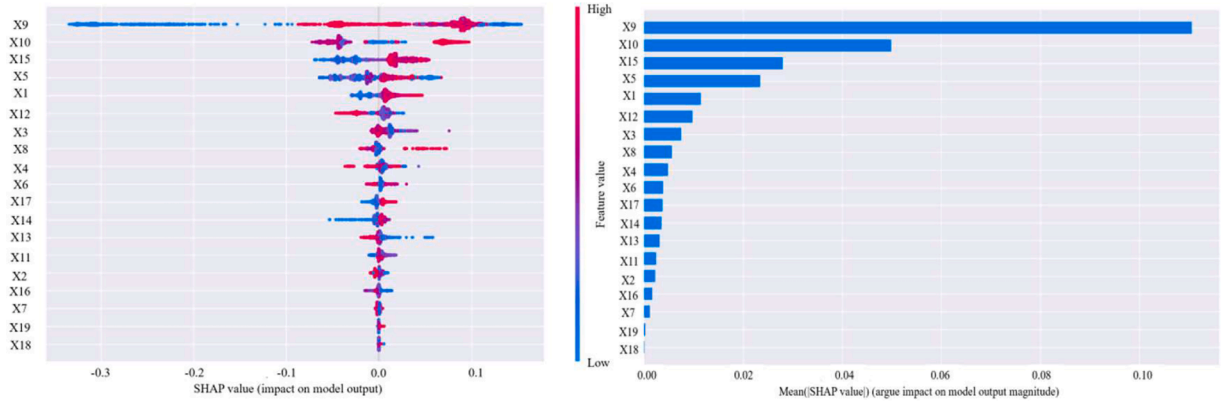
(1) The BO-CatBoost-MOEAD model proposed in this paper can effectively solve the large-diameter slurry SA multiobjective optimization problem. All six objectives of the SA are optimized. The optimized SA satisfies the constraints, which means that it does not exceed 80% of the specification requirements.
(2) The more adjusted large-diameter slurry shield construction parameters there are, the better the optimization of the large-diameter slurry SA. The best optimization rates for the five



(d)



(e)



(f)

Fig. 9. (*continued*).

Table 4
Optimization parameters for 5 scenarios.

Scenario	Input parameter
Scenario 1	X2, X4–X9
Scenario 2	X2, X4–X9, X10, X11
Scenario 3	X2, X4–X9, X10, X11, X3
Scenario 4	X2, X4–X9, X10, X11, X3, X17
Scenario 5	X2, X4–X9, X10, X11, X3, X17, X12

scenarios are 20.662%, 22.449%, 22.128%, 21.759%, and 23.436%, respectively, as shown in Table 6. The best optimization rate of the large-diameter slurry SA, 20.662%, occurs when only the cutterhead digging speed and propulsion pressure are adjusted in Scenario 1. The best optimization rate of the large SA is 22.449%, which occurs when the adjusted parameters of the large shield in Scenario 2 increase the slurry feed and slurry discharge densities. The largest number of large shield construction parameters adjusted in Scenario 5 correspond to the largest attitude optimization rate of 23.436%. Therefore, the

Table 5
SA input parameter ranges for silty fine sand.

Input parameter	Unit	Parameter range
X2	r/min	(0.9, 1.3)
X3	kN	(81,500, 94,800)
X4	kN	(6200, 12,000)
X5	kN	(6200, 12,000)
X6	kN	(13,600, 23,900)
X7	kN	(12,000, 15,200)
X8	kN	(12,000, 20,200)
X9	kN	(16,300, 26,800)
X10	t/m^3	(1.3, 1.5)
X11	t/m^3	(1.3, 1.5)
X12	m^3/h	(2140, 3030)
X17	Bar	(3.3, 3.4)

proposed BO-CatBoost-MOEAD model can effectively optimize large-diameter slurry SAs and offers a theoretical foundation for adjusting large-diameter slurry SAs in actual projects. The more parameters that are adjusted, the better the optimization effect of the large-diameter slurry SA is.

5. Discussion

5.1. Comparison of the predictive models

To obtain a prediction model with optimal accuracy, this paper uses the BO-BPNN, BO-SVM, BO-RF, BO-XGBoost and BO-CatBoost models to predict SA. The results of the SA predictions for the gravelly soil are shown in Fig. 10 and Table 7. The results of the SAs for the other two soils are shown in Fig. 11.

The BO-CatBoost-based SA prediction model has the best prediction results. The R^2 values are all above 0.9, the RMSE values are less than 1.4, and the MAE values are less than 0.6. The RMSE, MAE and R^2 of the CHHD (F1) dataset are 0.17, 0.128 and 0.987, respectively, for the angular gravelly soils and chalky clays. The RMSE, MAE and R^2 of the CHVD (F2) dataset are 0.111, 0.15 and 0.986, respectively. The RMSE,

Table 6
Comparison of SA optimization results for 5 scenarios.

Scenario		F1 (mm)	F2 (mm)	F3 (mm)	F4 (mm)	F5 (°)	F6 (°)	Average degree of optimization
Scenario 1	After optimization	18.356	21.811	34.166	38.951	28.627	3.604	20.662%
	Optimization percentage	53.436%	41.166%	15.016%	12.811%	5.875%	8.388%	
Scenario 2	After optimization	16.680	21.923	33.065	38.602	28.605	3.455	22.449%
	Optimization percentage	57.688%	40.863%	17.135%	13.591%	6.683%	12.182%	
Scenario 3	After optimization	17.212	21.653	33.241	38.932	28.589	3.466	22.128%
	Optimization percentage	56.338%	41.592%	16.797%	12.853%	6.703%	11.889%	
Scenario 4	After optimization	17.205	21.705	33.423	38.905	28.583	3.538	21.759
	Optimization percentage	56.355%	41.453%	16.446%	12.913%	6.764%	10.072%	
Scenario 5	After optimization	16.662	21.354	32.148	38.572	28.367	3.391	23.436%
	Optimization percentage	57.733%	42.399%	18.899%	13.659%	6.845%	13.803%	

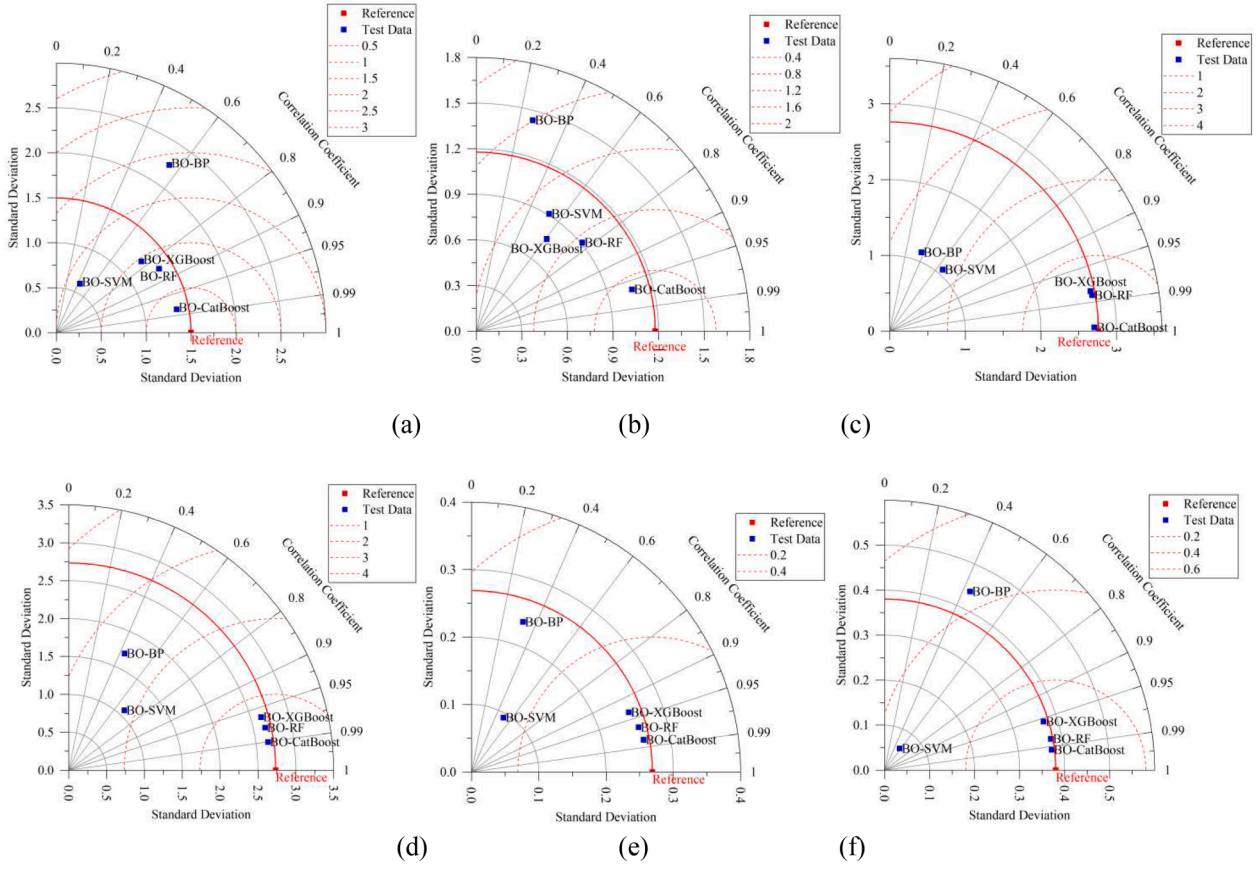


Fig. 10. Comparison of SA prediction results.

Table 7
Comparison of SA prediction results in gravel soil.

Forecast Objectives	Performance	BO-BP	BO-SVM	BO-RF	BO-XGBoost	BO-CatBoost
F1	RMSE	2.246	0.604	0.591	0.762	0.17
	MAE	2.078	0.735	0.4	0.572	0.128
	R^2	0.532	0.4402	0.844	0.74	0.987
F2	RMSE	1.435	0.908	0.576	0.74	0.111
	MAE	1.19	0.973	0.381	0.568	0.15
	R^2	0.211	0.511	0. 76	0.605	0.986
F3	RMSE	1.119	1.069	0.348	0.412	0.089
	MAE	1.043	0.946	0.213	0.294	0.065
	R^2	0.369	0.652	0.984	0.978	0.998
F4	RMSE	1.706	1.078	0.433	0.542	0.421
	MAE	1.485	1.016	0.295	0.403	0.309
	R^2	0.419	0.681	0.975	0.961	0.995
F5	RMSE	0.235	0.0935	0.052	0.069	0.013
	MAE	0.205	0.0893	0.025	0.042	0.01
	R^2	0.278	0.498	0.963	0.934	0.982
F6	RMSE	0.44	0.0586	0.051	0.079	0.013
	MAE	0.415	0.0502	0.028	0.052	0.01
	R^2	0.447	0.572	0.982	0.957	0.993

MAE and R^2 of the STHD (F3) dataset are 0.089, 0.065 and 0.998, respectively. The RMSE, MAE and R^2 of the STVD (F4) dataset are 0.421, 0.309 and 0.995, respectively. The RMSE, MAE and R^2 of the PA (F5) dataset are 0.013, 0.01 and 0.982, respectively, and the RMSE, MAE and R^2 of the HA (F6) dataset are 0.013, 0.01 and 0.993, respectively. The BO-CatBoost-based prediction model outperforms the other four models, indicating its superior fit in aligning the predicted and actual values for SA.

Fig. 11 illustrates the prediction results of these five prediction models for the other two soil types. The BO-SVM algorithm has the smallest RMSE and maintains a small prediction error under different soils and different prediction targets. The RMSE of the BO-BP algorithm is much larger than those of several other algorithms. The BO-CatBoost algorithm has the maximum prediction accuracy for different soils and different targets while ensuring a small prediction error. According to Fig. 10, BO-CatBoost has the highest fit and the smallest error for all three soil types. The R^2 values are all above 0.9, and the RMSEs are all less than 1.3. Therefore, BO-CatBoost can predict large-diameter slurry SAs more accurately.

5.2. Shield attitude control digital twin framework

The digital twin is used to open the channel from the physical system to the virtual system, as well as the virtual mapping of the physical system. In this paper, we emphasize big data analysis, so a digital twin framework for SA is constructed from three aspects: (1) the physical space of shield construction, (2) the information space, and (3) the interaction between the physical and information space. Fig. 12 shows the SA digital twin framework constructed in this paper.

(1) The physical space of shield construction includes a physical entity business layer and a data layer. The physical entity business layer extracts and expresses the processes, problems and knowledge of the entire shield construction process, constructs the business chain, and achieves perceptual access and fusion of multisource real-time data. The data layer utilizes collection, communication, digital integration and other technologies to manifest information, construct the whole process real-time big data chain, and achieve mapping from physical space to information space.
(2) The information space of shield construction includes a digital twin shell layer and an analysis layer. The digital twin shell layer utilizes fusion processing, node statutes, multisource heterogeneous data integration and conversion and other technologies to form a reusable digital chain, based on which the digital chain identifies, configures and manages digital twins. The digital twin analysis layer takes the big data analysis model as the core, integrates the knowledge model to achieve the intelligent analysis modeling of data–knowledge fusion, and achieves big data analysis of the shield construction attitude by combining the specific business requirements of SA monitoring, prediction and optimization.
(3) The interactive mapping of the physical and information spaces of the digital twin of the shield construction attitude achieves mutual mapping, timely interaction and efficient synergy through the integration of sensing, computing, communication, control and other information technologies and automatic control technologies. Additionally, it achieves on-demand responses, fast iteration and dynamic optimization through the process of entity voxelization and voxel materialization.

According to the SA adjustment control process and digital twin technology, an SA control twin system is designed to supplement and enhance the existing shield machine operating system, which primarily includes an attitude warning function and a parameter recommendation function. The interface design of the construction parameter recommendation value of the SA control system is shown in Fig. 13. The display area in the interface is divided into four main parts. The upper left side displays the actual and recommended values of the main thrust jack stroke in four directions. The lower left side displays the SA monitoring value, the four discs indicate the actual value of the current SA, and the lower four boxes display the SA monitoring value within 24 h. The upper right side shows the plan view of the construction section at

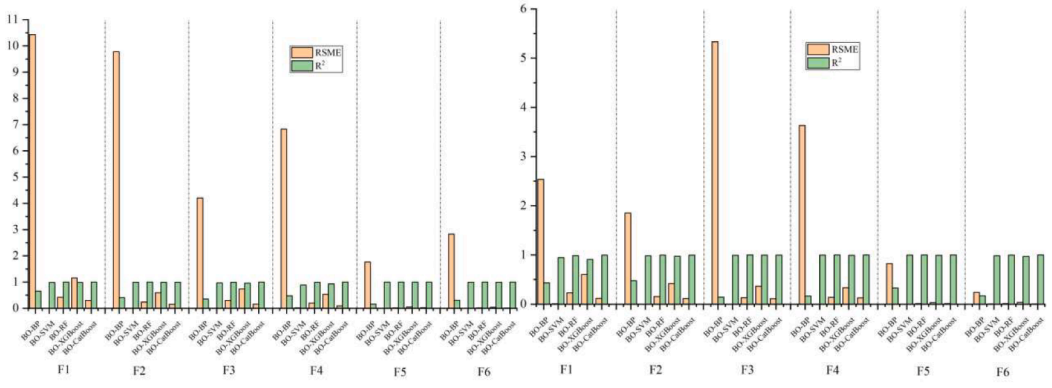


Fig. 11. Comparison of SA prediction results: (a) silty fine sand; (b) angular gravelly soils and chalky clays.

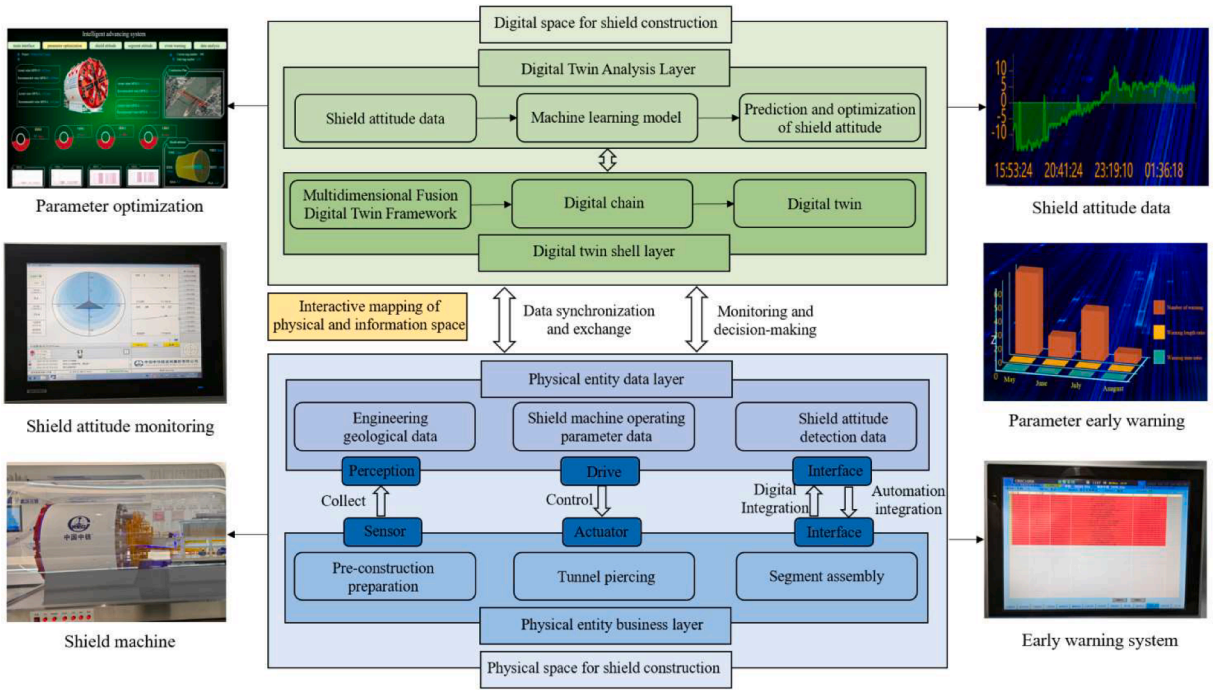


Fig. 12. Digital twin framework of SA.

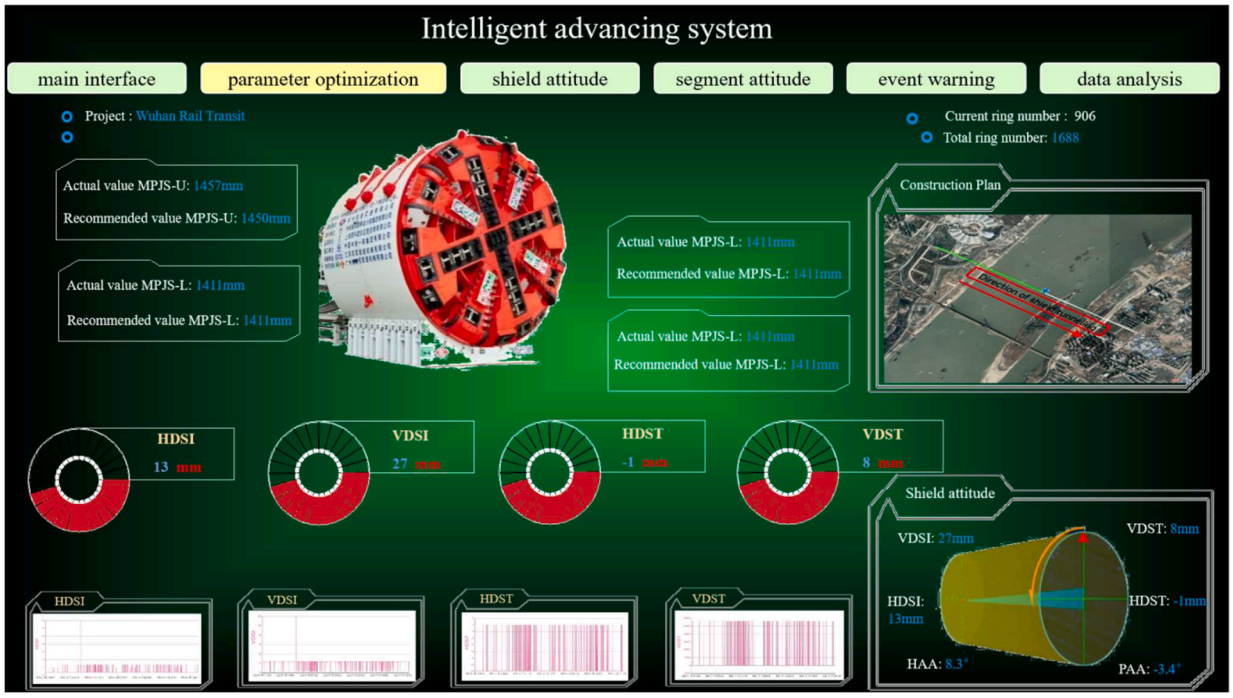


Fig. 13. SA control twin system.

that time. The lower right side displays the SA visualization, which provides the real-time values of the SA, with blue arrows indicating the direction of the shield when the CHHD and tail are rounded, red arrows indicating the direction of the shield when the CHVD and STVD are rounded, and orange arrows indicating the direction of the PA and HA when the PA and HA are rounded.

6. Conclusion and future work

In this study, a BO-CatBoost-MOEAD-based SA prediction and optimization model is established with 6 SA parameters and 19 influence parameters based on the shield part of the Wuhan subway project as an example. The BO-CatBoost algorithm is used to establish the nonlinear mapping correlation between the influencing factors and the prediction objectives, and it is used as the fitness function of the multiobjective optimization algorithm to improve the optimization efficiency of MOEAD multiobjective optimization. The shield construction parameters are adjusted according to the prediction and optimization results to control the SA and enhance shield construction safety and quality. In addition, a digital twin framework for SA is constructed to

better control the SA.

(1) BO-CatBoost is used to train and simulate the measured engineering data, and the obtained prediction model has higher accuracy and lower error than do the BPNN, SVM, RF, and XGBoost prediction models. The coefficients of determination of the six SA prediction targets are greater than 0.9, the RMSEs are less than 1.2, and the MAEs are less than 0.9 in all three soils. Hence, the BO-CatBoost model developed in this paper fit well, and the prediction results had small errors, indicating that this model can better predict the nonlinear relationships between the inputs and outputs.
(2) The SHAP analysis shows that the push group A–F thrust (X4–X9) has the most significant effect on the large-diameter slurry SA, followed by the cutter rotating speed (X2). Therefore, the main adjustment of these parameters can effectively control the SA.
(3) A multiobjective optimization method for large-diameter slurry SA based on BO-CatBoost-MOEAD is proposed. The more parameters that are adjusted, the better the optimization effect is. The maximum optimization rate can reach 23.436% compared with the original data. Moreover, a digital twin framework for intelligent prediction warning and SA optimization based on BO-CatBoost-MOEAD is proposed in this paper. The large-diameter slurry SA is predicted in advance based on existing data. Once the SA exceeds the warning range, the MOEAD optimization model is used to obtain the optimized shield construction parameters to adjust the SA.

Although the proposed BO-CatBoost-MOEAD model exhibits good prediction and optimization performance, it still has several limitations. The three types of soils used in this paper have relatively uniform distributions with little variation in soil properties; therefore, this model may not be effective in predicting and optimizing large-diameter slurry SAs for complex soils. This paper is based on the results of SHAP analysis to select several main parameters that have greater impacts on the large-diameter slurry SA to adjust and objectively optimize the large-diameter slurry SA. However, the penetration degree and tool size may affect the large-diameter slurry SA in the actual construction process. The prediction and optimization model of large-diameter slurry SA proposed in this paper needs to import real-time monitoring data manually; it cannot import monitoring data and realize prediction and optimization independently. Therefore, in future studies, a more comprehensive system of factors influencing large-diameter slurry SA will be established, and more shield projects will be adopted to increase the amount of data available for different soil types to improve the generalizability of the model. At the same time, the data monitoring platform will be connected to the proposed prediction and optimization model to realize independent prediction and optimization.

CRediT authorship contribution statement

Xianguo Wu: Writing – original draft, Conceptualization. **Jingyi Wang:** Formal analysis, Data curation. **Zongbao Feng:** Investigation. **Hongyu Chen:** Resources, Methodology. **Tiejun Li:** Validation, Supervision. **Yang Liu:** Writing – review & editing, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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